

**An autonomous car detects roads, pedestrians,
and signs.**

Explain:

- A. Vision pipeline**
- B. Camera calibration**
- C. Depth estimation**

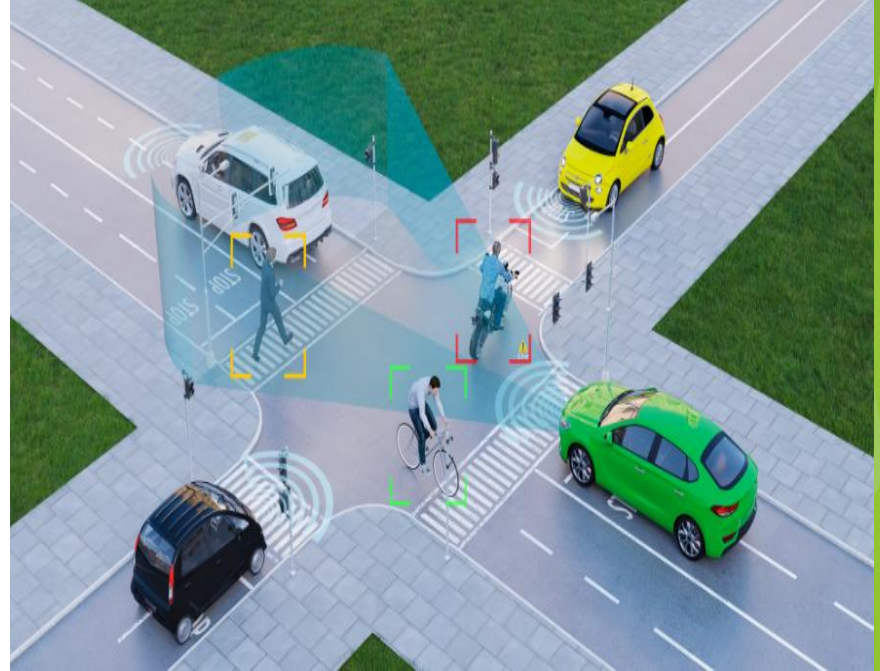
**Presented by:
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1 - Introduction to Autonomous Vehicle Perception

An autonomous car relies on a sophisticated vision pipeline to perceive its environment accurately.

This perception system detects roads, pedestrians, and traffic signs to ensure safe navigation.

Understanding the vision pipeline is essential for developing reliable autonomous driving systems.



2 - A. Vision Pipeline Overview

The vision pipeline processes raw sensor data, primarily from cameras, to interpret the surroundings.

It includes stages such as image acquisition, preprocessing, object detection, and classification.

The pipeline enables the vehicle to identify and understand key elements like roads, pedestrians, and signs in real-time.



3 - B. Image Preprocessing and Feature Extraction

Raw images are enhanced through preprocessing techniques like normalization and noise reduction.

Feature extraction algorithms identify edges, textures, and shapes relevant to objects in the scene.

These features form the basis for subsequent detection and classification tasks.

e) Given below is the designed pipeline for autonomous driving system:

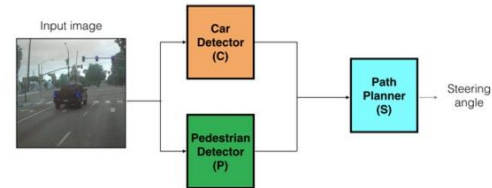


Figure 1: The input camera image is given to two modules: the Car Detector C and the Pedestrian Detector P. C outputs a set of bounding boxes localizing the cars. P outputs a set of bounding boxes localizing the pedestrians. The bounding boxes are then given to a Path Planner S which outputs the steering angle. Assume all these submodules are supervised learning algorithms.

- What data do you need to train the submodules in the pipeline presented in Figure 1?
(3 marks)
- Explain how would you collect the data for the 3 modules; (X_c, Y_c) , (X_p, Y_p) and (X_s, Y_s) , referring to Car Detector, Pedestrian Detector and Path Planner respectively.
(6 marks)
- After setting up a camera on the hood of your car, an expert driver in your team has collected the dataset A in a city where most roads are straight. She always drives in the center of the lane. Describe two problems that can occur if you train a model on A and test it on real-world data.
(4 marks)

4 - C. Object Detection and Classification

Advanced deep learning models, such as convolutional neural networks (CNNs), are employed to detect objects.

The system classifies objects into categories like pedestrians, stop signs, or traffic lights.

Accurate detection and classification enable the vehicle to make informed driving decisions.



5 - Summary and Importance

The vision pipeline is a critical component that allows autonomous cars to perceive their environment reliably.

Proper implementation of preprocessing, detection, and classification stages ensures safety and efficiency.

Continuous improvements in vision algorithms are vital for advancing autonomous driving capabilities.

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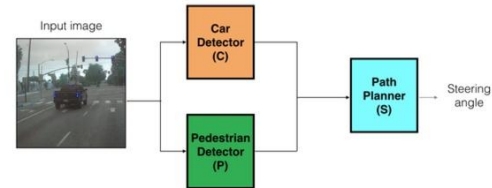


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